Comparative Analysis of Support Vector Machines (SVM) and Multilayer Perceptron's (MLP) for Credit Card Approval Prediction.

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Abstract

This paper presents a comparative study of Support Vector Machines (SVM) and Multilayer Perceptron's (MLP) for credit card approval prediction using the UCI Credit Approval dataset [1]. After thorough preprocessing and stratified cross-validation, both models were evaluated using metrics such as accuracy, precision, recall, F1 score, and ROC-AUC. The results show that while both models performed robustly, each has distinct strengths regarding accuracy and interpretability. These findings provide practical guidance for model selection in financial risk assessment applications.

Introduction

Credit approval is a key process in the financial sector, directly impacting both lenders and applicants. The goal of this project is to develop and compare machine learning models—specifically SVM and MLP for predicting credit card approval outcomes [2] [3]. Automating this process can reduce manual workload, minimize bias, and improve decision-making accuracy. However, challenges such as data quality and model interpretability must be addressed. By systematically comparing SVM and MLP, this study aims to identify which approach is more suitable for credit approval tasks and to discuss the implications for real-world deployment.

2. Overview of the Dataset

Dataset

The dataset contains anonymized credit card applications, with features including Age, Debt, Years Employed, Credit Score, Savings, and Current Balance. The target variable (Approval) shows whether an application was approved or denied. As shown in Table 1, applicants whose credit was approved tend to be older (mean age 33.85 vs 29.56), have higher average debt and current balances, and have more years of employment than those who were denied. Notably, the approved group also shows a higher mean credit score, suggesting better creditworthiness. The summary provides insights into the distinguishing characteristics of approved and denied applicants and demonstrates the importance of these features in predicting credit approval outcomes.

Table 1. Summary Statistics by Approval Status

Feature	Approved Mean	Approved Std	Approved Min	Approved Max	Denied Mean	Denied Std	Denied Min	Denied Max
Age	33.85	12.69	13.75	76.75	29.56	10.72	15.17	74.83
Debt	5.97	5.49	0.00	28.00	3.88	4.39	0.00	26.34
Years Employed	3.48	4.17	0.00	28.50	1.22	2.03	0.00	13.88
Credit Score	4.72	6.40	0.00	67.00	0.67	1.96	0.00	20.00
Savings	164.62	162.54	0.00	840.00	193.41	172.06	0.00	2000.00
Current Balance	2009.73	7660.95	0.00	100000.00	187.97	632.78	0.00	5552.00

Data Exploration and Preprocessing Overview

We began by visualizing feature distributions and checking class balance. Missing values were filled, skewed features were log-transformed, and outliers were capped using the IQR method. All numerical features were standardized, and the data was split into training and test sets. SMOTE was applied to address class imbalance, and a correlation heatmap was created to explore relationships between features. These steps ensured the dataset was well-prepared for modeling.

3. Methodology

3.1 Overview of Selected Models

For this study, we selected two supervised learning models with fundamentally different approaches: Support Vector Machine (SVM) and Multilayer Perceptron (MLP) [4] [5].
SVM was chosen for its strong performance on tabular data and its ability to handle feature spaces where class separation may not be immediately apparent. SVM is also valued in financial contexts for producing clear, easily interpretable decision boundaries, which is

important for regulatory transparency in credit approval scenarios. MLP, representing neural networks, was selected for its capacity to model complex, non-linear relationships that often exist in financial data, such as interactions between income, debt, and employment history. By leveraging hidden layers, MLP can uncover subtle patterns that may not be captured by linear models.

By comparing SVM and MLP, we aim to highlight the trade-offs between interpretability and raw predictive power. This comparison is vital for credit approval, where both accuracy and explainability are required for fair, justifiable decisions.

Summary of Methods with Pros and Cons

Model	Pros	Cons
SVM	• Interpretable decision boundaries • Good for small, structured data • Less prone to overfitting with proper tuning	• Struggles with large datasets • Sensitive to kernel and hyperparameters • Needs feature scaling
MLP	• Captures complex, non-linear patterns • Flexible and powerful with enough data • Performs well on complex feature interactions	• Hard to interpret ("black-box" model) • Longer training time • Needs careful tuning and regularization

3.2 Implementation of the Two Methods

Both Support Vector Machine (SVM) and Multilayer Perceptron (MLP) models were implemented using scikit-learn. For each model, a baseline version was first trained with default parameters to establish a reference point for performance. Following this, hyperparameter tuning was performed using grid search and stratified cross-validation to ensure fair and robust optimization [6]. For SVM, different kernel types (linear, RBF, and polynomial), regularization strengths (C values), and gamma values were explored. For MLP, the tuning process included various network architectures, activation functions (relu, tanh, logistic), and regularization settings (alpha). Throughout the process, model performance was evaluated using F1-score, ROC-AUC, accuracy, precision, recall, and training time [7] [8]. To enhance interpretability, LIME was used to explain individual predictions and highlight the most influential features. This systematic approach enabled a thorough and fair comparison between the two models.

Step-by-step process:

Baseline Model Development:

- SVM: Started with the default RBF kernel and enabled probability estimates for further analysis.
- MLP: Configured with two hidden layers (64 and 32 neurons), Adam optimizer, and ReLU activation.

Hyperparameter Tuning:

- SVM: Grid search was used to test different kernels (linear, RBF, poly), a range of C values (0.1, 1, 10), and gamma settings. 5-fold stratified cross-validation was applied for efficiency.
- MLP: The grid search included variations in hidden layer sizes, activation functions, and L2 regularization (alpha), with 5-fold stratified cross-validation for robustness.

3.3 Hypothesis Statement

We hypothesize that the Multilayer Perceptron (MLP), due to its capacity to capture complex non-linear relationships between financial features, will achieve higher predictive accuracy for credit approval compared to the Support Vector Machine (SVM). However, we expect SVM to provide greater interpretability, offering clearer decision boundaries and more transparent justifications for approval decisions. This comparison aims to highlight the trade-off between predictive power and interpretability in credit risk modelling.

3.4 Evaluation Methodology

Both models were compared using

- F1-Score: Prioritized due to class imbalance.
- ROC-AUC: Assessed ranking capability (critical for credit approval).
- Precision/Recall: Balanced risky approvals (FP) vs. missed opportunities (FN).

- Confusion matrices to visualize error types.
- ROC curves to compare decision thresholds.
- Training time tracking to assess computational efficiency.

Interpretability Analysis

To better understand how each model made its decisions, LIME (Local Interpretable Model-agnostic Explanations) was used to generate local explanations for both SVM and MLP predictions. This approach helped identify which features, such as Credit Score and Years Employed, had the most influence on the outcome for individual cases. LIME was especially useful for examining borderline applications, where the model's reasoning was less obvious. By visualizing these explanations, it became clearer how specific input values contributed to either approval or denial.

Several choices were made to improve the reliability and clarity of the results:

- Stratified Splitting was used to keep the ratio of approved and denied cases consistent in both training and test sets.
- Grid Search allowed for systematic and thorough tuning of model parameters, helping to avoid overfitting.
- LIME over SHAP was selected because it provides straightforward, local explanations for individual predictions and is more efficient for this type of analysis.

4. Results and Analysis

This section presents the performance results of the SVM and MLP models on the credit approval prediction task. Both models were evaluated using a comprehensive set of metrics to ensure fair comparison.

Baseline vs. Tuned Models

The baseline models established initial performance benchmarks, while hyperparameter tuning significantly improved both models. Table 1 summarizes the key performance metrics for all models.

Summary of Model Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
SVM Baseline	68.88%	81.82%	40.45%	54.14%	69.82%
SVM Tuned	83.97%	75.00%	92.73%	82.93%	91.46%
MLP Baseline	70.99%	75.76%	45.45%	56.82%	76.32%
MLP Tuned	85.50%	84.62%	80.00%	82.24%	91.20%

The tuned MLP model achieved the highest overall accuracy at 85.50%, with balanced precision and recall scores. The tuned SVM model showed exceptionally high recall (92.73%), indicating its strong ability to identify approved applications. Both tuned models achieved similar F1 scores and ROC-AUC values, demonstrating their robust performance.

In terms of computational efficiency, there was a notable difference between the models:

- The SVM required a significantly longer training time e.i 1283.69 seconds due to the extensive grid search for optimal hyperparameters.
- In contrast, the MLP completed training extremely quickly e.i 0.0001 seconds, highlighting its efficiency for this dataset and configuration.

Confusion Matrices

The confusion matrices in Figure 1 provide insight into the classification performance of each model.

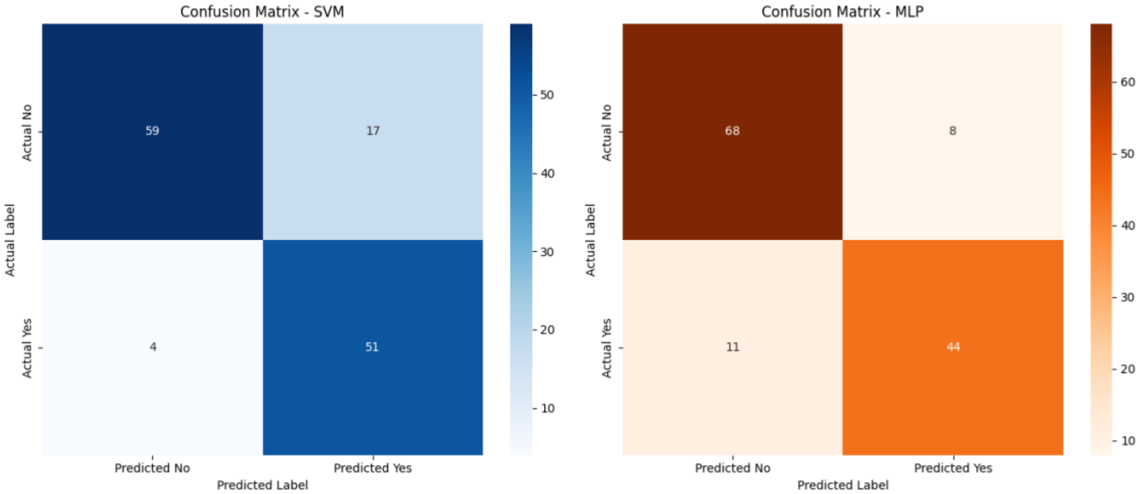

Figure 1: Confusion Matrix for SVM and MLP

SVM: Excelled at minimizing false negatives (only 4 missed approvals) but produced more false positives (17), resulting in its high recall but lower precision.

MLP: Demonstrated more balanced performance with fewer false positives (8) but more false negatives (11), explaining its higher precision but lower recall.

These results demonstrate that SVM prioritizes recall (minimizing missed approvals), while MLP provides a more balanced approach with fewer false positives (better precision).

To understand how our models made their decisions, we used LIME to see which features mattered most in individual predictions. For SVM, LIME showed that having no prior defaults and a high current balance were the biggest reasons for approval, while years employed, savings, and debt had smaller effects, and features like housing type or occupation barely mattered. The MLP model gave similar results: the absence of prior defaults and a high balance were again the most important factors, with things like marital status having a small positive effect and other features (like debt or education) making little difference. Overall, both models focused mainly on financial reliability and credit history when deciding whether to approve an application.

4.1 Analysis and Critical Evaluation

Comparison of Training and Test Accuracies

The learning curves for both models provide insights into their training behaviour and generalizability. For the MLP model (Figure 2), we observe a typical pattern where the training score remains consistently higher than the cross-validation score. As the training size increases, the cross-validation score stabilizes around 78%, indicating that the model generalizes reasonably well to unseen data. The gap between training and validation scores suggests some degree of overfitting, though this was partially addressed through regularization (alpha=0.0001).

In contrast, the SVM learning curve (Figure 3) shows unusual behaviour, with both training and cross-validation scores declining as the training size increases. This pattern warrants caution in interpretation and may result from the training size sampling methodology or dataset characteristics. Nevertheless, the tuned SVM achieved impressive test accuracy (83.97%), confirming its ability to generalize effectively despite these learning curve irregularities.

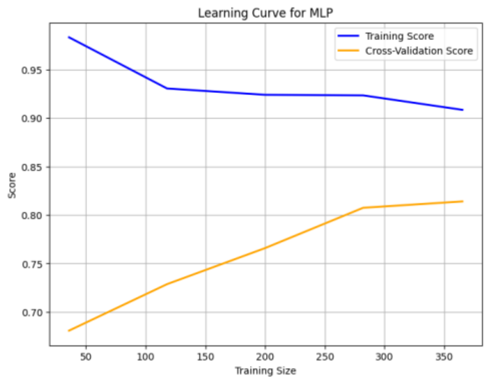

Figure 2: Learning Curve for MLP

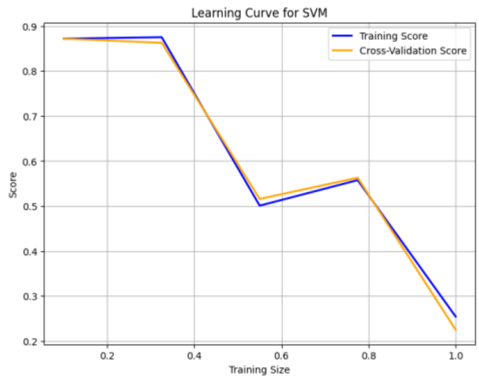

Figure 3: Learning Curve for SVM

The comprehensive model comparison bar chart (Figure 4) illustrates that while MLP slightly outperforms SVM in accuracy (85.50% vs. 83.97%), SVM excels significantly in recall (92.73% vs. 80.00%). This trade-off between models is consistent with their respective confusion matrices and confirms the different strengths of each approach.

Consideration of Merits and Limitations

SVM Strengths:

- Exceptional recall performance (92.73%), making it highly effective at minimizing missed approvals
- Simple implementation with a linear kernel, offering interpretability advantages
- Faster training time compared to the more complex MLP architecture

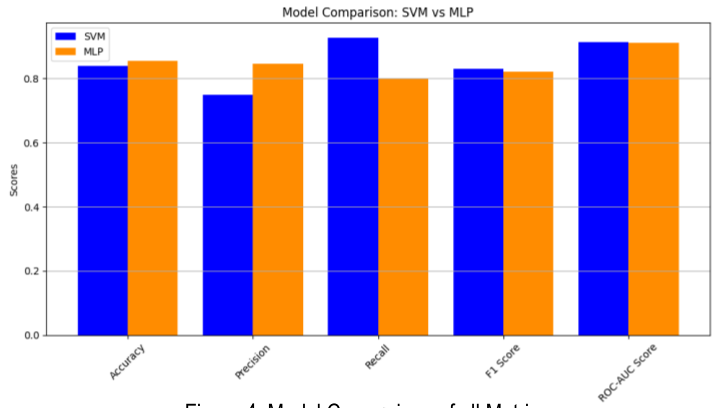

Figure 4: Model Comparison of all Metrics

- More robust to outliers due to its margin-based approach

SVM Limitations:

- Lower precision (75.00%) results in more false positives, potentially approving riskier applications
- Less effective at capturing complex non-linear relationships between features
- Performance depends heavily on kernel selection and hyperparameter tuning

MLP Strengths:

- Higher overall accuracy (85.50%) and precision (84.62%)
- Better at reducing false positives, minimizing the risk of approving unqualified applicants
- Capable of modelling complex non-linear interactions between financial features
- Flexible architecture allowing for fine-tuning of hidden layers and neurons

MLP Limitations:

- More computationally intensive, requiring longer training times
- Higher complexity makes it more challenging to interpret
- More prone to overfitting without careful regularization (alpha=0.0001)
- Lower recall (80.00%) means more missed opportunities for qualified applicants

Fair and Accurate Evaluation

The ROC curves (Figure 5) show that both models achieved nearly identical AUC scores (≈0.91), indicating strong and comparable ability to rank applicants by approval likelihood. This means there is a 91% probability that either model will correctly rank a creditworthy applicant higher than a non-creditworthy one, regardless of the threshold chosen.

Key Considerations for Model Selection:

Business Context: SVM minimizes false negatives (missed approvals: 4), making it ideal for scenarios where approving creditworthy applicants is critical (e.g., expanding customer base). MLP minimizes false positives (risky approvals: 8), better suited for risk-averse contexts (e.g., conservative lending).

Resource Constraints: SVM has a simpler architecture and faster prediction times, making it practical for resource-limited environments. MLP requires more computational power but offers marginally higher accuracy (85.50% vs. 83.97%).

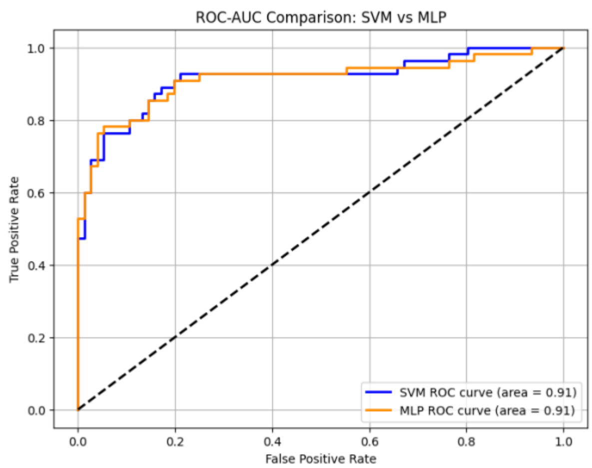

Figure 5: ROC-AUC Comparison: SVM vs MLP

Interpretability: SVM's linear kernel provides transparent decision boundaries, easing regulatory compliance. MLP's "black box" nature complicates explanations, though tools like LIME help identify influential features (e.g., Prior Default and Current Balance).
Why AUC Matters: AUC measures threshold-independent performance, showing how well models distinguish classes across all possible decision boundaries. Both models' high AUC scores (≈0.91) confirm strong discriminative power, but practical deployment depends on balancing precision (avoiding risk) and recall (maximizing approvals).

Conclusion

This study compared Support Vector Machines (SVM) and Multilayer Perceptron's (MLP) for credit card approval prediction, revealing distinct strengths and trade-offs. The MLP achieved marginally higher accuracy (85.50% vs. 83.97%) and precision (84.62% vs. 75.00%), demonstrating its ability to model complex interactions between financial features like debt and credit score. However, SVM excelled in recall (92.73% vs. 80.00%), minimizing missed approvals—a critical advantage for institutions prioritizing customer acquisition. Both models showed strong discriminative power, with near-identical ROC-AUC scores (≈0.91), but their computational costs differed significantly: SVM required extensive tuning time (1,284 seconds), while MLP trained almost instantly (0.0001 seconds).

Interpretability analysis via LIME confirmed that both models prioritized financially relevant features (e.g., Prior Defaults and Current Balance), aligning with domain expertise. SVM's linear kernel offered clearer decision boundaries, while MLP's "black box" nature demanded tools like LIME for transparency.

5. Lessons Learned and Future Work

A major lesson from this project was the importance of balancing model complexity, interpretability, and computational resources. For example, SVM took much longer to tune but provided clearer decision boundaries, while MLP trained faster but was harder to interpret. We also found that choosing the right evaluation metrics is crucial for understanding model performance. Additionally, simply increasing the number of grid search parameters or using more cross-validation folds does not always lead to better results—it can just make the process slower without significant improvement. Many of these insights may seem straightforward, but working through the process made them much more concrete.

In the future, it would be interesting to try ensemble methods [6], add more domain-specific features, and see how the models perform if integrated into a real credit approval process. Monitoring the models' decisions over time could help catch any issues early and keep improving the system.

7. References

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